

Progression of the Rim Fire—Yosemite, California

The winter of 2012–2013 was among the driest on record for California, setting the stage for an active fire season in the summer of 2013. At the time, the Rim Fire was the third largest in California since record-keeping began in 1932.

The VIIRS DNB on the Suomi NPP satellite tracked the growth of the fire between August 20 and September 4, 2013. The brightest, most intense parts of the fire glow white. Pale gray smoke streams away, generally to the north. Thin clouds obscured the view on some days, particularly August 31, September 1, and September 3. On August 20, the Moon was full, so the landscape reflected a large amount of moonlight. The background grew progressively darker as the new Moon approached on September 5.

The perimeter of the fire changed along different fronts from day to day, depending on winds and firefighting efforts. On August 24, firefighters focused on containing the western edge of the fire to prevent it from burning into Tuolumne City and the populated Highway 108 corridor. They also fought the eastern edge of the fire to protect Yosemite National Park (outlined in yellow). These efforts are evident in the images: Between August 23 and 24, the eastern edge held steady, the western edge receded, and the fire grew in the southeast.

On the morning of August 25, 2013, fire managers reported that the blaze was growing in the north and east. With the fire burning aggressively and moving east into Yosemite, August 26 and 27 proved challenging days for firefighters. But over the next few days, they began to gain control after a series of burnout operations along the fire's northern and eastern edges. On August 29, the evacuation advisory for Tuolumne City was lifted. The southeastern flank continued to burn intensely into the first week of September.



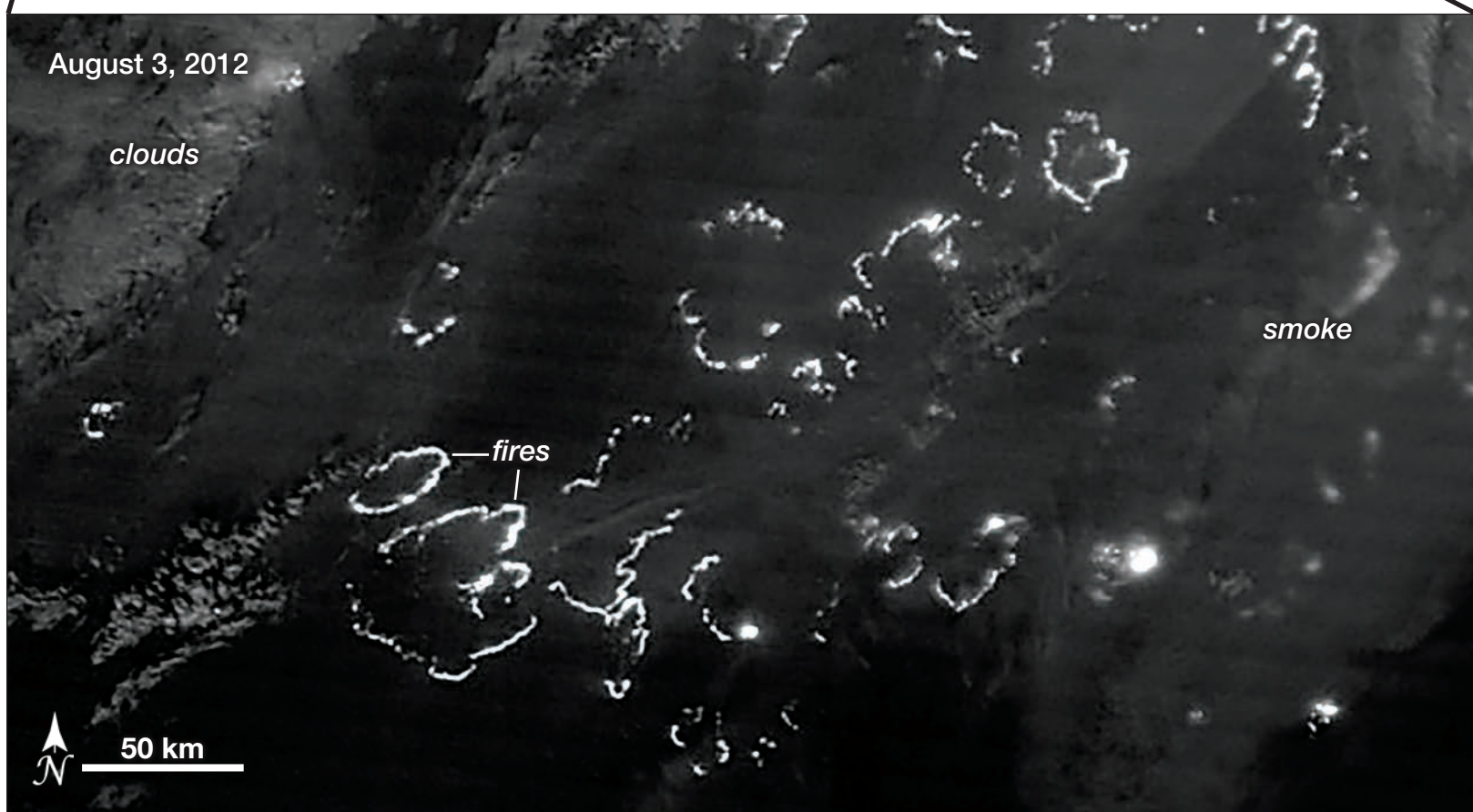
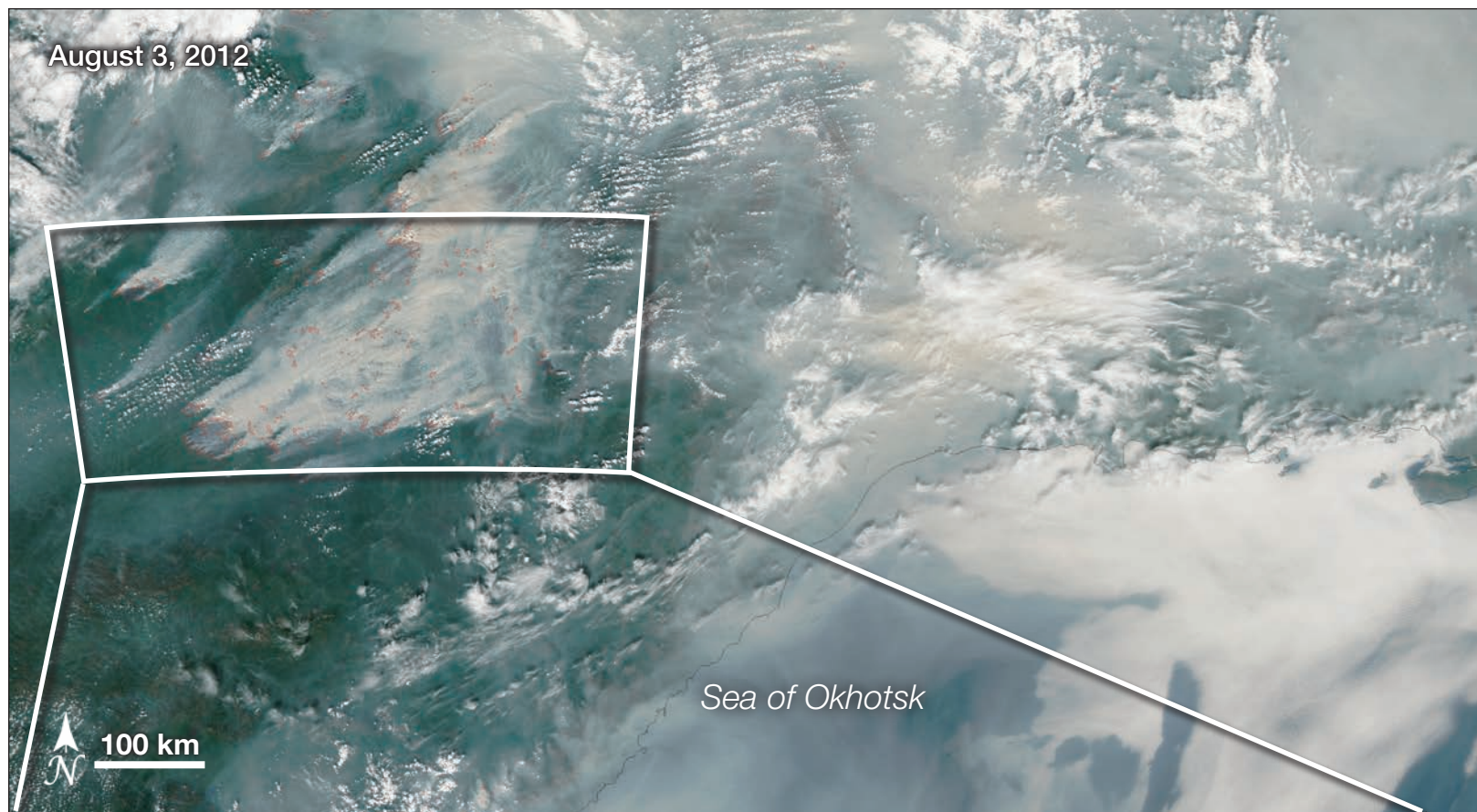


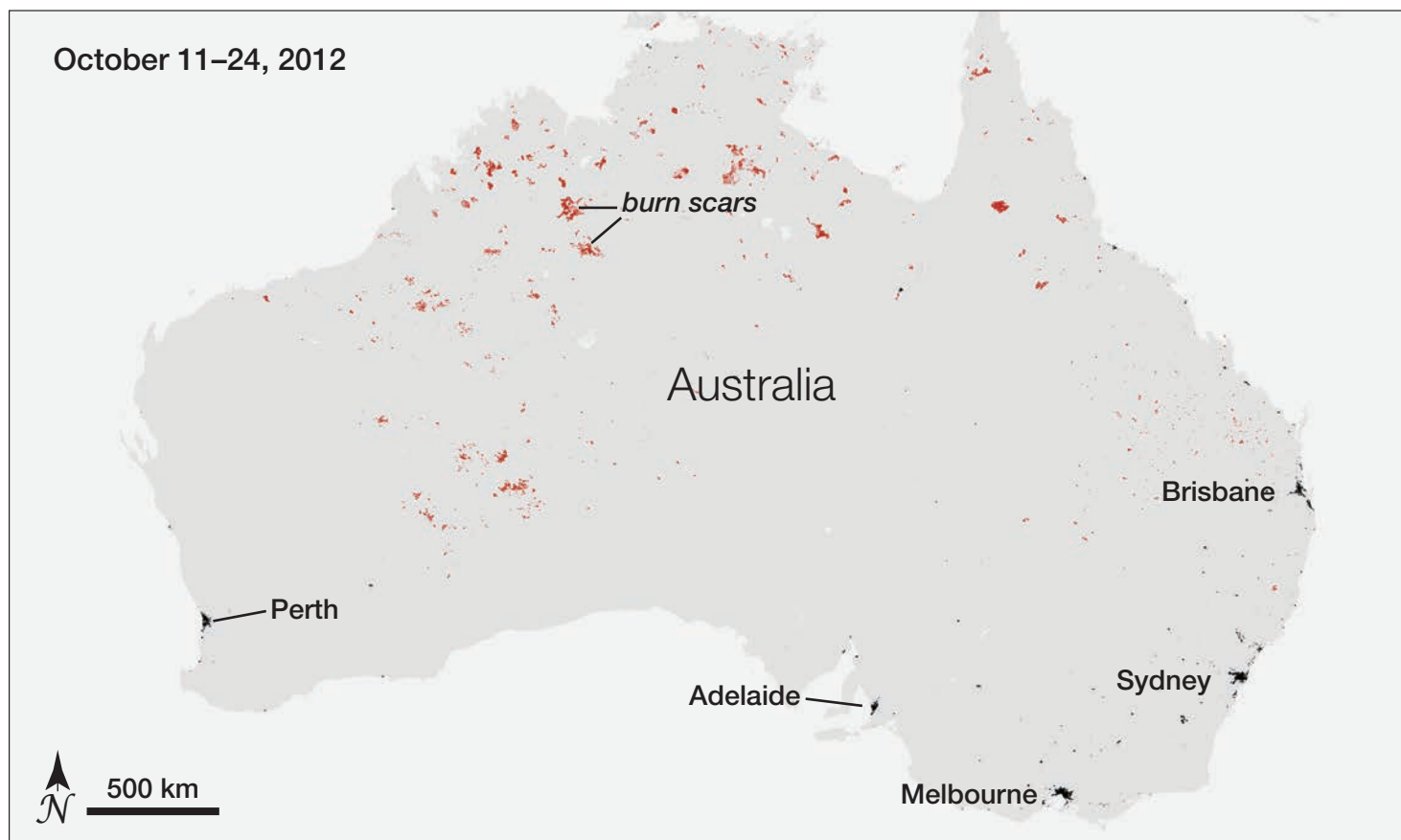
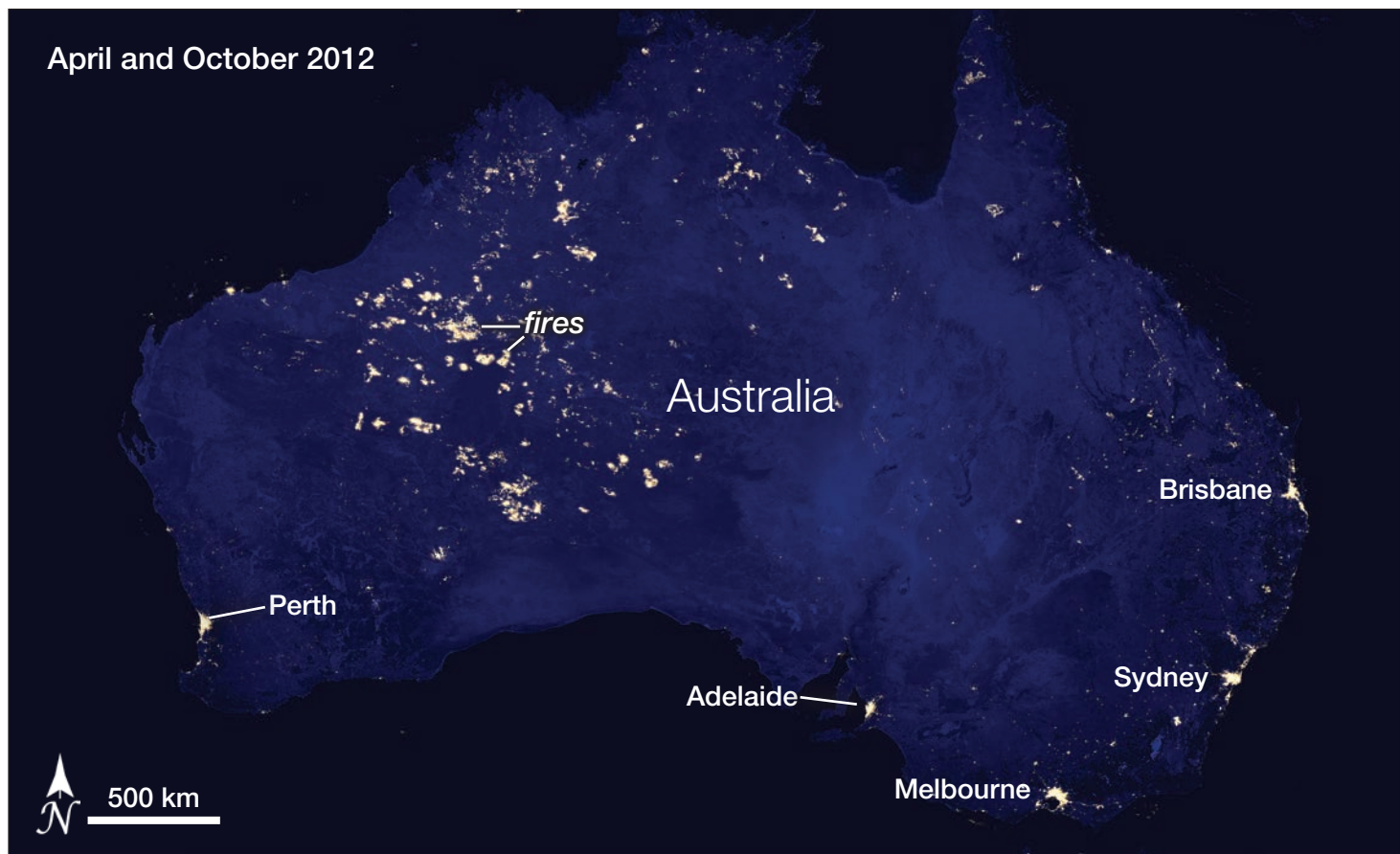
Spreading Like Wildfire—Yakutia, Siberia, Russia

On August 3, 2012, the MODIS instrument on NASA’s Aqua satellite acquired a daytime view of a group of wildfires blazing in eastern Siberia, as shown in the top panel. The fires detailed in the white box were burning in the southeastern part of Yakutia, near the port city of Yakutsk along the Lena River. Red dots indicate hot spots where MODIS detected unusually warm surface temperatures associated with fires. Smoke is visible emanating from the hot spots.

The VIIRS DNB on Suomi NPP acquired a nighttime image of that same group of fires on the same date, as seen in the bottom panel. In this image, the brightest fires are white; smoke is light gray.

The vast majority of Russian wildfires occur in Siberia, generally along the southern border with China and Mongolia.







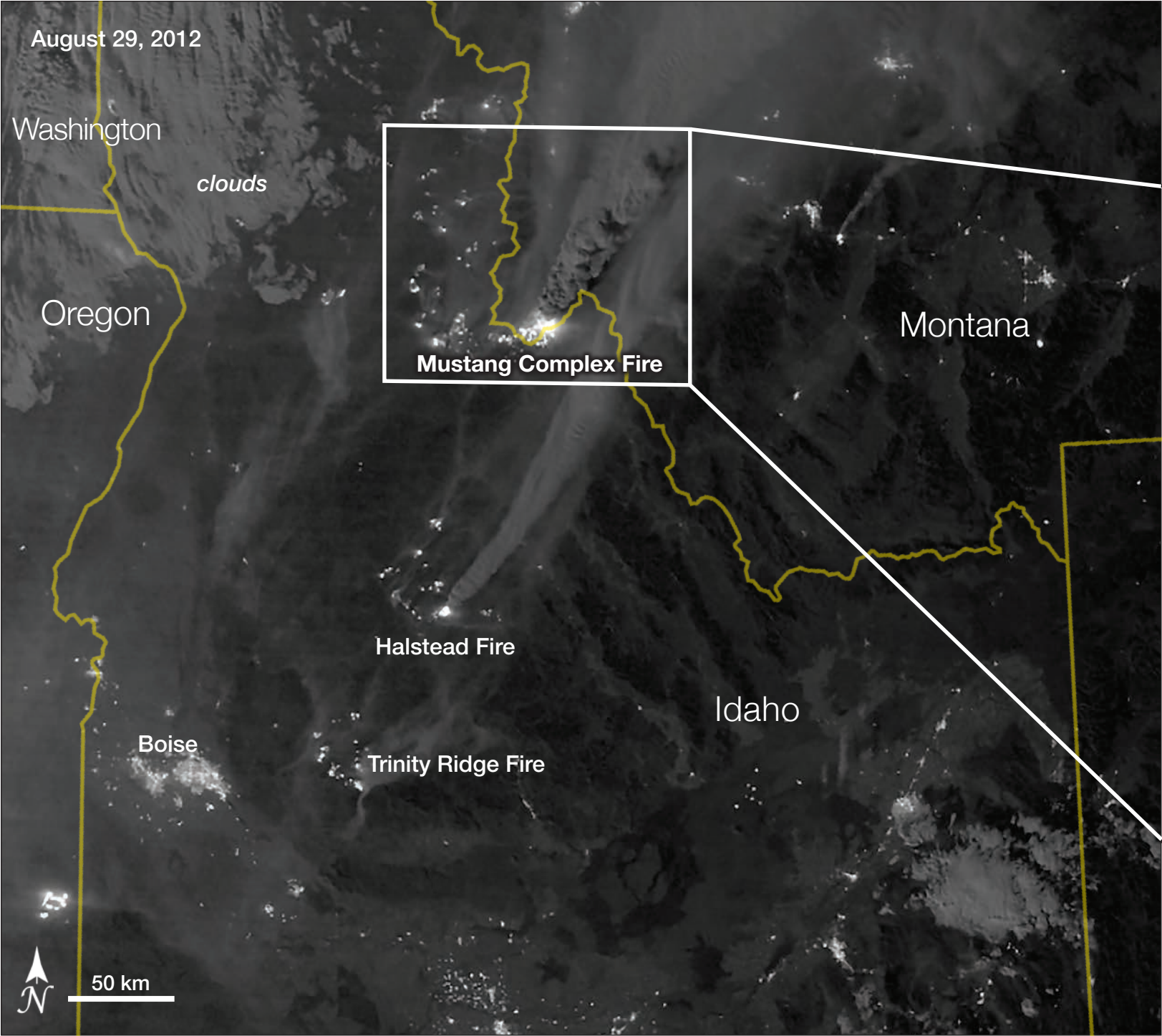
City Lights, or Not—Australia

The top image at left shows the nightlights of Australia as observed by the VIIRS DNB on the Suomi NPP satellite in April and October 2012. The composite image includes manmade light sources and the light from wildfires.

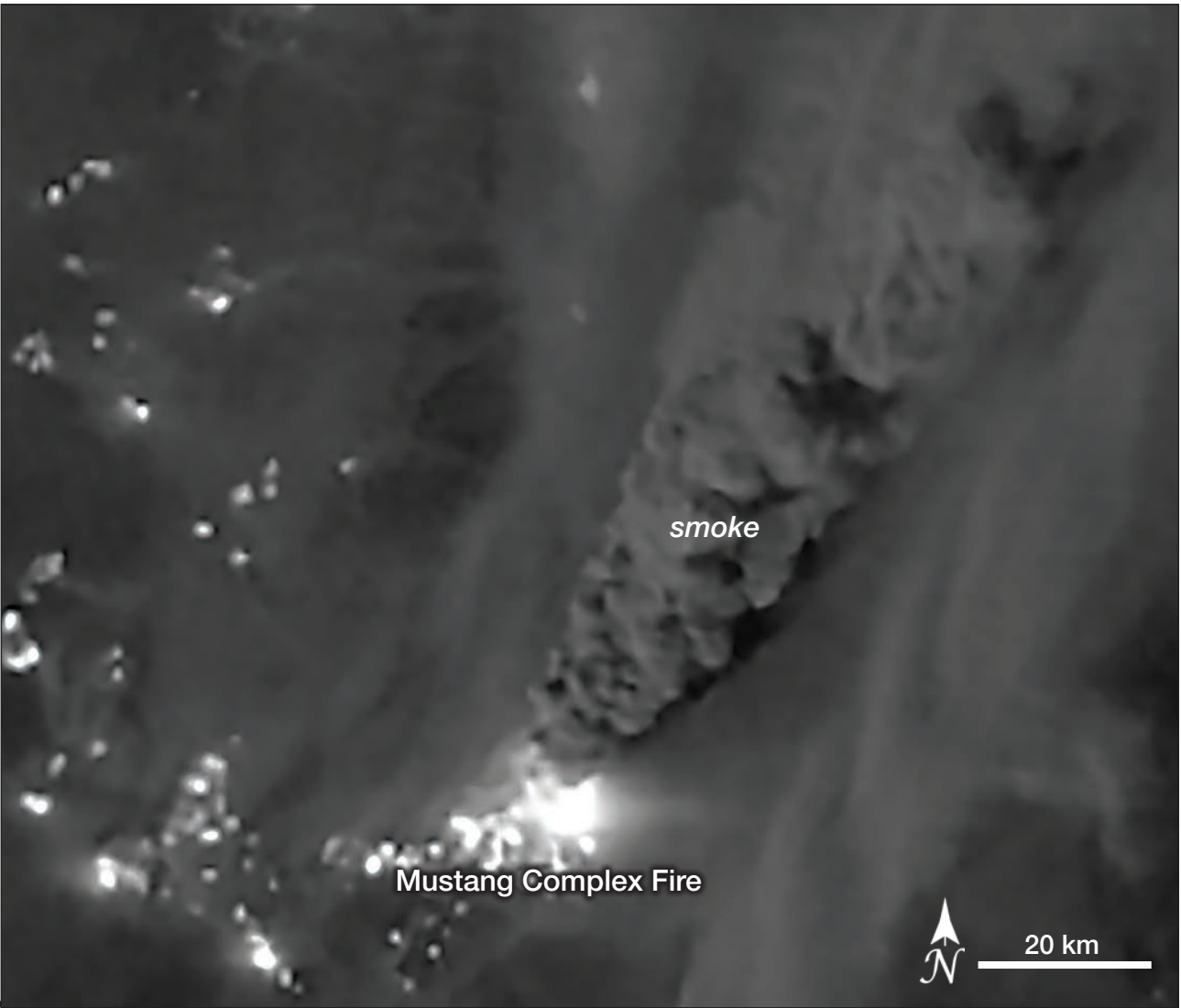
The lower map is a mosaic showing the burned areas of the landscape (red) from October 11–24, 2012, combined with urban areas (black). The data were collected by the MODIS instruments on NASA's Terra and Aqua satellites. In effect, the lower map shows where fires burned that month. Though many rural areas of interior Australia are dry and relatively barren by some standards, there is still enough vegetation to burn. Not every light in the night view matches up with a fire—partly because the fire map does not include fires from April and partly because not every fire leaves a scar that is detectable from space. Even simple cloud cover could prevent burn scars from being observed. Aside from the fires, some of the nightlights appearing in uninhabited areas can be attributed to lightning, flares of natural gas and other activities from oil drilling or mining operations, and fishing boats—all of which can show up as points of light.

Billowing Smoke in the Night—Idaho and Montana

On August 29, 2012, the VIIRS DNB on the Suomi NPP satellite captured this nighttime view of wildfires burning in Idaho and Montana. When the image was acquired, the Moon was in its waxing gibbous phase (i.e., it was more than half lit but less than full). Numerous hot spots from the Mustang Complex Fire are visible in northern



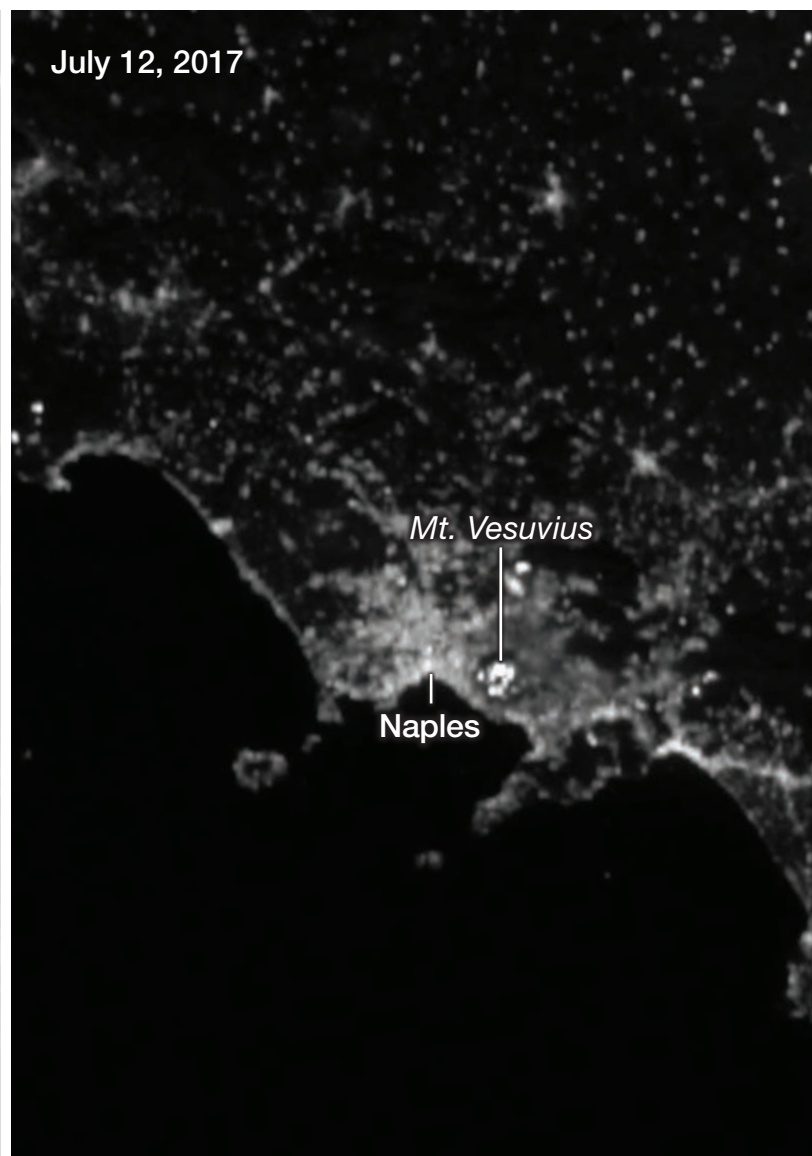
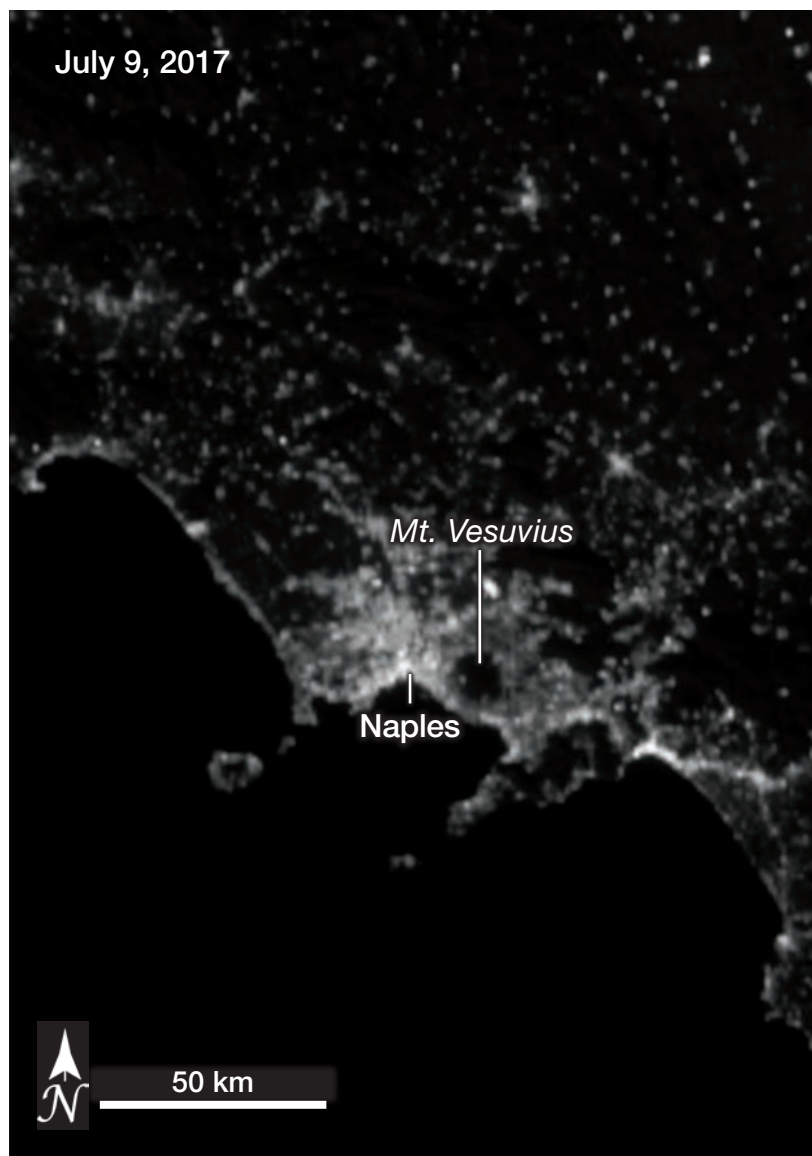
Idaho. A plume of thick, billowing smoke streams west from the brightest fires near the Idaho–Montana border. The Halstead and Trinity Ridge fires are visible to the south. In addition to the fires, city lights from Boise and other smaller cities appear throughout the image. A bank of clouds is located west of the Mustang Complex, over southeastern Washington and northeastern Oregon.





Fires Light Up Mount Vesuvius—Naples, Italy

The forest preserve on Mount Vesuvius normally keeps the Italian mountain shrouded in darkness when viewed in nighttime photographs by astronauts or in satellite images of Naples. However, that was not the case on July 12, 2017, when the VIIRS DNB on Suomi NPP captured an image (below, right) of wildfires lighting up the slopes of the volcano. For comparison, the July 9, 2017, image (below, left) was taken before the fires began. The July fires burned much of the woodlands in Vesuvius National Park, which was established in 1995. The park protects more than 600



types of plants, 100 species of birds, and many small mammals and reptiles. Mount Vesuvius is best known for a catastrophic eruption in 79 A.D. that destroyed the cities of Pompeii and Herculaneum. Its last major eruption occurred in 1944.

For comparison, the MODIS instrument captured this daylight image (below) of the fires (red dots) on July 12. Note how the extent of Naples and the area around the volcano are less clearly defined.





Volcanoes

The Infrared Glows of Kilauea's Lava Flows—Hawaii

In early May 2018, an eruption on Hawaii's Kilauea volcano began to unfold. The eruption took a dangerous turn on May 3, 2018, when new fissures opened in the residential neighborhood of Leilani Estates. During the summer-long eruptive event, other fissures emerged along the East Rift Zone. Lava from vents along the rift zone flowed downslope, reaching the ocean in several areas, and filling in Kapoho Bay.

A time series of Landsat 8 imagery shows the progression of the lava flows from May 16 to August 13. The night view combines thermal, shortwave infrared, and near-infrared wavelengths to tease out the very hot lava (bright white), cooling lava (red), and lava flows obstructed by clouds (purple).